

Java Truth Tables for AND (&&), OR (||), and NOT (!)

Learning Objectives

After this topic, you will be able to:

- Understand how logical operators work.
 - Read and use truth tables.
 - Predict the result of logical expressions.
 - Apply logical operators in if conditions.
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1. What is a Truth Table?

A **truth table** shows the result of a logical expression for every possible combination of boolean values (true or false).

Java has three main logical operators:

- && (AND)
 - || (OR)
 - ! (NOT)
-

2. AND (&&) Truth Table

The **AND** operator returns true **only if both conditions are true**.

A	B	A && B
true	true	true
true	false	false
false	true	false
false	false	false

Example

```
boolean isLoggedIn = true;  
boolean isVerified = true;  
  
System.out.println(isLoggedIn && isVerified);
```

Output

```
true
```

3. OR (||) Truth Table

The **OR** operator returns true if **at least one condition is true**.

A	B	A B
true	true	true
true	false	true
false	true	true
false	false	false

Example

```
boolean hasCoupon = false;  
boolean isPremium = true;  
  
System.out.println(hasCoupon || isPremium);
```

Output

```
true
```

4. NOT (!) Truth Table

The **NOT** operator reverses a boolean value.

A	!A
true	false
false	true

Example

```
boolean accountBlocked = false;  
  
System.out.println(!accountBlocked);
```

Output

```
true
```

5. Real-World Examples

Login Access (AND)

```
if (isLoggedIn && isVerified) {  
    System.out.println("Access Granted");  
}
```

Scholarship Eligibility (OR)

```
if (cgpa >= 3.5 || sportsQuota) {  
    System.out.println("Eligible");  
}
```

Account Check (NOT)

```
if (!accountBlocked) {  
    System.out.println("User can login");  
}
```

6. Common Mistakes

✗ Confusing && with ||.

- && → **Both conditions must be true.**
- || → **At least one condition must be true.**

✗ Forgetting that ! simply reverses true and false.

7. Interview Questions

Q1. When does && return true?

Answer: Only when **both conditions are true.**

Q2. When does || return false?

Answer: Only when **both conditions are false.**

Q3. What is the output of !true?

Answer:

false

Q4. Which operator is used to reverse a boolean value?

Answer: ! (NOT)

8. Output-Based Questions

Question 1

```
System.out.println(true && false);
```

Output

false

Question 2

```
System.out.println(true || false);
```

Output

true

Question 3

```
System.out.println(!false);
```

Output

```
true
```

Cheat Sheet

Operator	Meaning	Returns true When...
&&	AND	Both conditions are true
,		,
!	NOT	Reverses the boolean value

Remember

- && → **Both**
- || → **At least One**
- ! → **Reverse**